# 3.Data Splitting and Cross Validation

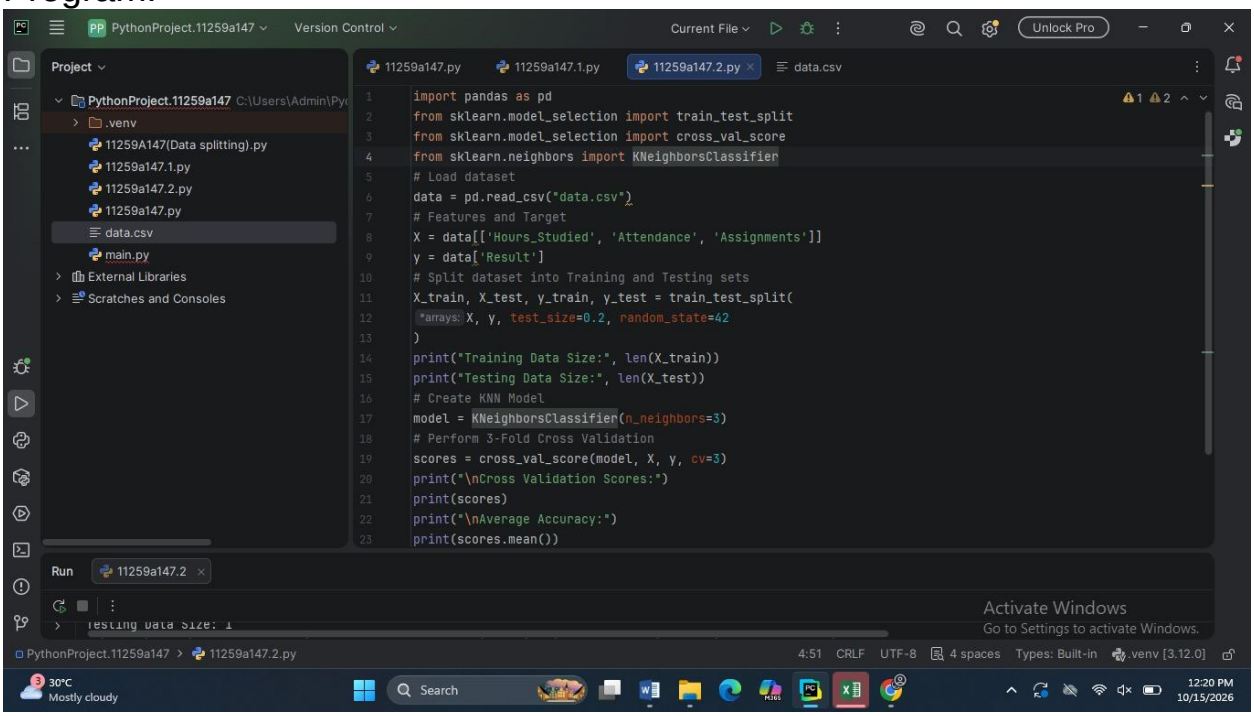
## Aim

To write a program that splits the given dataset into training and testing sets and then applies k-fold cross validation, so that the model's performance is checked on different parts of the data and not on a single split.

## Algorithm

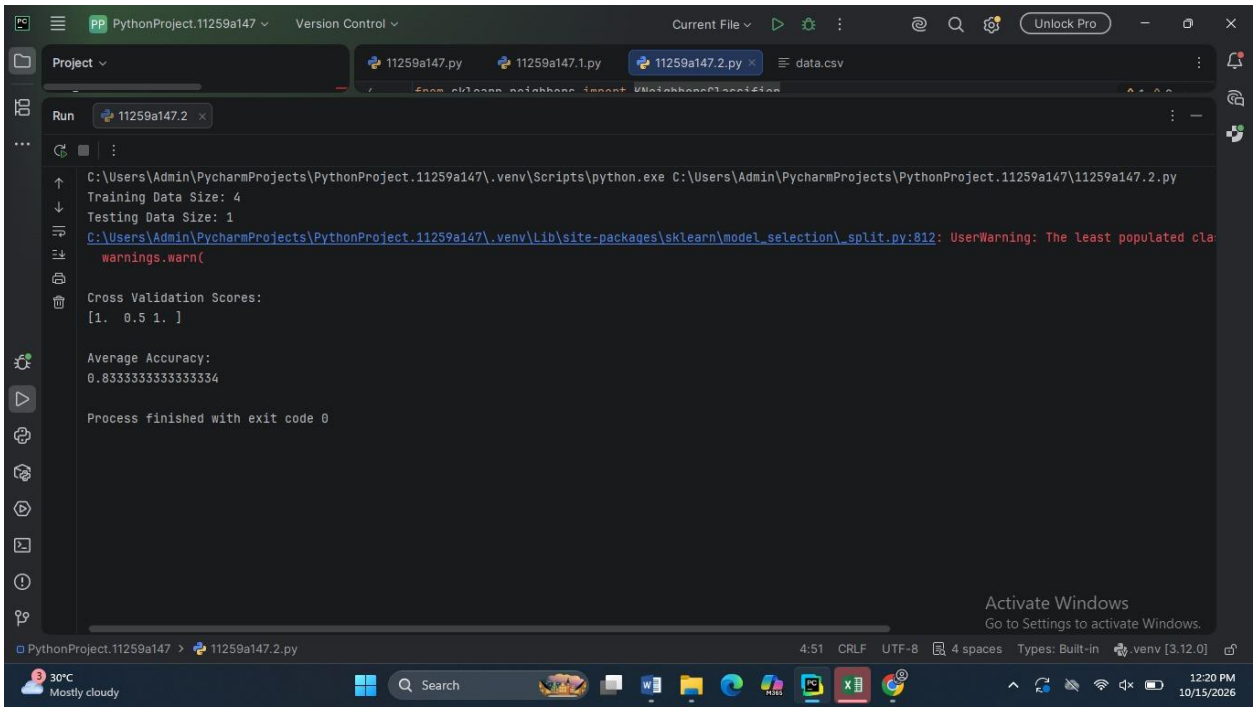
1. Start the program.
2. Import the required libraries (pandas, train_test_split, KFold or cross_val_score from sklearn.model_selection, and a model such as LogisticRegression).
3. Load the dataset from the input file.
4. Separate the data into input features (X) and the target variable (y ).
5. Split the data into training and testing sets using train_test_split (for example 80% training and 20 % testing), with a fixed random_state.
6. Display the number of records in the training and testing sets.
7. Choose the number of folds k (for example k = 5) and create the k-fold object.
8. Divide the training data into k equal folds.
9. For each fold, use that fold as the validation set and the remaining k-1 folds as the training set.
10. Train the model on the training folds and calculate its accuracy on the validation fold.
11. Repeat steps 9 and 10 until every fold has been used once for validation.
12. Calculate the mean of the k accuracy scores to get the cross-validation score.
13. Display the score of each fold and the mean score.
14. Stop the program.

Program:



```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.model_selection import cross_val_score
from sklearn.neighbors import KNeighborsClassifier
# Load dataset
data = pd.read_csv("data.csv")
# Features and Target
X = data[['Hours_Studied', 'Attendance', 'Assignments']]
y = data['Result']
# Split dataset into Training and Testing sets
X_train, X_test, y_train, y_test = train_test_split(
    *arrays: X, y, test_size=0.2, random_state=42
)
print("Training Data Size:", len(X_train))
print("Testing Data Size:", len(X_test))
# Create KNN Model
model = KNeighborsClassifier(n_neighbors=3)
# Perform 3-Fold Cross Validation
scores = cross_val_score(model, X, y, cv=3)
print("\nCross Validation Scores:")
print(scores)
print("\nAverage Accuracy:")
print(scores.mean())
```

Result:



```
C:\Users\Admin\PycharmProjects\PythonProject.11259a147\.venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\PythonProject.11259a147\11259a147.2.py
Training Data Size: 4
Testing Data Size: 1
C:\Users\Admin\PycharmProjects\PythonProject.11259a147\.venv\Lib\site-packages\sklearn\model_selection\_split.py:812: UserWarning: The least populated cla
  warnings.warn(

Cross Validation Scores:
[1.  0.5 1. ]

Average Accuracy:
0.8333333333333334

Process finished with exit code 0
```